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Product Backlog: Turn-Based RPG (BenLeViRPG)

Idea: a small turn-based RPG. The player picks a character and then fights a series of enemies one after the other. Each turn the player chooses what to do (attack, special attack or heal), then the enemy plays. The enemies get stronger each time and the last fight is against a boss.

Backlog

#	Name	How to Demo	Notes
1	Choose a character	Launch the app, a list of characters is shown with their stats. Click one, it is selected and the game starts.	Each character has different HP, attack and defense values.
2	Show the battle screen	Start a game, the player and the enemy are displayed with their name, HP and the buttons for the possible actions.	Built with Swing. The screen updates after every action.
3	Attack	During a battle, click Attack; the enemy's HP goes down.	Damage is based on the player's attack and the enemy's defense. HP can't go below 0.
4	Enemy turn	Do any action; after that the enemy attacks by itself and the player's HP goes down.	The enemy picks its action randomly.
5	Special attack	Click Special, the enemy takes more damage than with a normal attack.	Limited to 3 uses per game. Can trigger a critical hit.
6	Heal	Lose some HP, then click Heal; the player's HP goes up.	Can't go above max HP. Limited to 3 heals per game.
7	Win / lose	Bring the enemy's HP to 0, a victory message appears. If the player's HP reaches 0, a game over message appears.	Action buttons are disabled once the battle is over.
8	Next enemy	Win a battle, a new enemy appears that is stronger than the previous one.	Enemies have different stats and types.
9	Final boss	Beat all the normal enemies, the boss fight starts. Beat the boss, a "you win" screen is shown.	The boss has higher stats and a more powerful special attack.
10	Restart	At the end of the game (win or lose) click Restart; we go back to character selection.	Everything is reset: player, enemies, progress.
11	XP & level up	Win a battle; the hero's XP bar increases. Gain enough XP; the hero levels up and their stats (HP, ATK, DEF) increase.	XP gained depends on the enemy's level vs the hero's level. Minimum 25 XP per win. Each level-up raises stats and max HP.

12	Replay battle	Lose a battle; a "Play again" button appears. Click it; the hero's HP is restored and the same enemy reappears.	Allows retrying the current fight without resetting the whole game. Hero keeps their level and stats.
13	UI polish	Launch the app; the hero selection and battle screen display character sprites, colored type names, and styled stat boxes. Colors match the game's visual identity.	Warm dark palette, gold buttons, character sprites loaded from resources, HTML labels for colored type names.
14	Critical hit	Fight any enemy; occasionally a "crits!" message appears in the log and the damage dealt is higher than usual.	Each character has a crit chance (e.g. 5%). A crit multiplies damage by 2. Applies to normal and special attacks.
15	Type system	Pick a character with a type advantage against the first enemy (e.g. Water vs Fire); the damage dealt is visibly higher than against a neutral enemy.	4 types: Normal, Fire, Water, Thunder. Multipliers: x1.3 (advantage) / x0.8 (disadvantage).
16	Coin system & shop	Win a battle; coins are earned and displayed. Open the shop and buy a stat upgrade; the hero's stats increase.	Coins are earned after each battle. The shop offers stat upgrades (ATK, DEF, HP).
17	JUnit tests	Run mvn test; all tests pass.	Tests cover damage calculation, healing, type multipliers, level-up, win/lose conditions and special attacks.

Scope

We implemented 3 characters (Warrior, Archer, Beast), 3 normal enemies and 1 boss, with 3 actions (attack, special, heal). The type system and critical hits were added as part of the advanced topics.

Advanced Topics

1. Unit Testing with JUnit — deterministic unit tests using JUnit 5 covering all game logic.
2. UX & Game Design — type system design, game balance, visual feedback (sprite images, colored type names,status boxes).